The one-loop potential along the abelian valley of the torus is periodic, reproduces C133's linear term at the origin, and saturates at order 1/L

For $SU(2)$ Yang–Mills on T^3_L with a constant abelian background $a = a^3_i T^3$, the zero-point energy of all modes, relative to $a = 0$, is the exact periodic function

$$U(a) = \frac{2\hbar c}{\pi^2 L}\,\Phi(La), \qquad \Phi(b) = \sum_{m\in\mathbb{Z}^3\setminus\{0\}} \frac{1-\cos(m\cdot b)}{|m|^4} \;\geq 0,$$

with a of dimension inverse length ($[A] = L^{-1}$ in the convention of G07 Proposition 7), obtained by Poisson summation of the charged frequencies $\hbar c|k \pm a|$ over $k \in (2\pi/L)\mathbb{Z}^3$; the divergent part of the sum is a-independent under any periodic regularization and drops out. Near the origin $\Phi(b) = \pi^2|b| + \frac{1}{6}C_M|b|^2 + O(|b|^3)$ with $C_M = \lim_{b\to 0}[\sum'_m e^{im\cdot b}|m|^{-2} - 2\pi^2/|b|]$, so $U(a) = 2\hbar c|a| + \frac{C_M}{3\pi^2}\hbar c L|a|^2 + \cdots$. The linear term $2\hbar c|a|$ is exactly the $k = 0$ contribution, four oscillators of angular frequency $c|a|$ with zero-point energy $\frac{1}{2}\hbar c|a|$ each, which are the transverse oscillators of C133 along its abelian valley: the field-theory computation reproduces the mechanism of G07 and G08 term by term. The nonzero modes contribute the correction $U_\perp = U - 2\hbar c|a| = O(\hbar c L|a|^2)$, of order $g^{4/3}\hbar c/L$ in the rescaled variables of the Feshbach note, which is the input (H3) needs, with the sign of C_M (negative for the cubic lattice) deciding that the correction lowers the valley potential. Globally U is bounded by $2\hbar c \max\Phi/(\pi^2 L) = O(\hbar c/L)$ and vanishes exactly at $La \in 2\pi\mathbb{Z}^3$, the holonomies that are central in the fundamental representation: the confinement of the constant modes along the valley is linear only for $La \ll 1$, saturates at order $\hbar c/L$, and has degenerate minima at the centre elements, which is the small-volume origin of the electric-flux sectors. Since the zero-mode gap is $g^{2/3}\hbar c/L \ll \hbar c/L$, the ground state is localized at $La \sim g^{2/3}$ and never reaches the saturation; the crossover $z \simeq 2$ is where it does. Sources: this is the spatial-torus form of Weiss's finite-temperature potential (Phys. Rev. D 24 (1981) 475, metadata), and it appears in the small-volume literature (Lüscher 1983, abstract); the derivation below is self-contained. Nothing here is promoted.

1. Frequencies in a constant abelian background

Let $a_i = \alpha_i T^3$, $i = 1, 2, 3$, with $\vec{\alpha} \in \mathbb{R}^3$, and write a for $\vec{\alpha}$ when no confusion arises. Decompose the adjoint fluctuation $\tilde{A}_i = \tilde{A}^3_i T^3 + \tilde{A}^+_i T^+ + \tilde{A}^-_i T^-$. The covariant derivative $D_j = \partial_j + [a_j, \cdot]$ acts on the neutral component as ∂_j and on the charged components as $\partial_j \pm i\alpha_j$, so on the Fourier mode $e^{ik\cdot x}$, $k \in (2\pi/L)\mathbb{Z}^3$, the charged components have covariant momentum $k \pm a$. The quadratic form $\frac{1}{2g^2}\int |D_i\tilde{A}_j - D_j\tilde{A}_i|^2$ is, for a constant abelian background, the free transverse form with k replaced by $k \pm a$ on the charged components, so after the usual transverse gauge choice the frequencies are

$$\hbar\omega^{(0)}_k = \hbar c|k| \quad \text{(neutral, two polarizations)},$$

$$\hbar\omega^{(\pm)}_k = \hbar c|k \pm a| \quad \text{(charged, two polarizations each)},$$

for $k \neq 0$. At $k = 0$ the neutral components are the valley directions themselves and the charged components of $\tilde{A}_j$ have frequency $|a|$ for each of the two charges and, since there is no transversality at $k = 0$, for the components of $\tilde{A}_j$ orthogonal to the valley in field space: with $a = \alpha_1 T^3$ along x_1 only, these are the charged parts of $\tilde{A}_2, \tilde{A}_3$, four real oscillators of angular frequency $c|a|$, while the charged parts of $\tilde{A}_1$ are gauge directions of frequency zero removed by Gauss's law.

The $k = 0$ term is C133. The SU(2) matrix model of C133 with $\vec{x}_1 = \lambda\hat{n}$, $\vec{x}_2 = \vec{x}_3 = 0$ has potential $\frac{1}{2}g^2(|\vec{x}_1 \times \vec{x}_2|^2 + |\vec{x}_1 \times \vec{x}_3|^2 + \cdots)$, whose transverse Hessian at that point gives the

components of $\vec{x}_2, \vec{x}_3$ orthogonal to $\hat{n}$, four oscillators of frequency $g|\lambda|/\sqrt{m}$; in the field-theory variables of G07 Proposition 7 ($m = \hbar L^3/(g^2 c)$, $g_B^2 = \hbar c L^3/g^2$) this frequency is $c|a|$. The zero-point energy $4 \cdot \frac{1}{2}\hbar c|a| = 2\hbar c|a|$ is the confining term of G07 Theorem 6(1) and the floor of G08 Theorem 2 along the valley.

2. Poisson summation

The zero-point energy of the fluctuations relative to $a = 0$, summing two polarizations and both charges, is

$$U(a) = \frac{\hbar c}{2} \sum_{k \in (2\pi/L)\mathbb{Z}^3} \sum_{\text{pol}=1,2} \big(|k+a| + |k-a| - 2|k|\big) = \hbar c \sum_{k \in (2\pi/L)\mathbb{Z}^3} \big(|k+a| + |k-a| - 2|k|\big),$$

where the $k = 0$ term is $2\hbar c|a|$ and the neutral modes cancel. Each term is nonnegative by convexity of $|\cdot|$ and of order $|a|^2/|k|$, so the sum diverges in three dimensions; under a periodic regularization (lattice dispersion ω_{lat} with the exact momentum grid) the divergent part is a-independent, because $\sum_{k \in \text{grid}} \omega_{\text{lat}}(k+a)$ is a periodic function of a with period $2\pi/L$ whose mean over a period is the a-independent Brillouin-zone integral. The a-dependent part is obtained by Poisson summation. For $f(k) = |k|$ the distributional Fourier transform is $\hat{f}(x) = \int |k| e^{-ik\cdot x} d^3k = -8\pi/|x|^4$ for $x \neq 0$ (the formula $\widehat{|k|^\alpha} = 2^{\alpha+3}\pi^{3/2}\frac{\Gamma((3+\alpha)/2)}{\Gamma(-\alpha/2)}|x|^{-3-\alpha}$ at $\alpha = 1$, with $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$), and

$$\sum_{k \in (2\pi/L)\mathbb{Z}^3} f(k+a) = \Big(\frac{L}{2\pi}\Big)^3 \sum_{m \in \mathbb{Z}^3} \hat{f}(mL)\, e^{im\cdot La}$$

with the $m = 0$ term the divergent, a-independent one. Hence

$$U(a) = \hbar c\Big(\frac{L}{2\pi}\Big)^3 \sum_{m \neq 0} \Big(-\frac{8\pi}{L^4|m|^4}\Big)\big(e^{im\cdot La} + e^{-im\cdot La} - 2\big) = \frac{2\hbar c}{\pi^2 L} \sum_{m \neq 0} \frac{1 - \cos(m \cdot La)}{|m|^4} = \frac{2\hbar c}{\pi^2 L}\Phi(La).$$

Lattice corrections change $\hat{f}$ only at $|x| \lesssim a_{\text{lat}}$, while the $m \neq 0$ terms live at $|x| = |m|L \geq L$, so they are $O(a_{\text{lat}}^2/L^2)$ relative.

Properties of Φ: it is nonnegative, periodic with period 2π in each component, invariant under the cubic group, bounded by $2\sum' |m|^{-4}$, and zero exactly on $2\pi\mathbb{Z}^3$.

3. Expansion at the origin and the sign of the correction

Φ is not smooth at 0. Its Laplacian is the periodic Green's function $G(b) = \sum'_m e^{im\cdot b}|m|^{-2}$, which has the Coulomb singularity $2\pi^2/|b|$ at the origin (from $\int d^3m\, e^{im\cdot b}|m|^{-2}$) and a finite remainder: $G(b) = 2\pi^2/|b| + C_M + O(|b|^2)$, where C_M is the constant term of the periodic Coulomb Green's function of the cubic lattice, the analytically continued Epstein sum $\sum'_m |m|^{-2}$. Since $\Delta(\pi^2|b|) = 2\pi^2/|b|$, the function $\Phi - \pi^2|b|$ has Laplacian $C_M + O(|b|^2)$ near 0, and the only quadratic form with cubic symmetry and Laplacian C_M is $\frac{1}{6}C_M|b|^2$; so

$$\Phi(b) = \pi^2|b| + \tfrac{1}{6}C_M|b|^2 + O(|b|^3), \qquad U(a) = 2\hbar c|a| + \frac{C_M}{3\pi^2}\hbar c L|a|^2 + O(\hbar c L^2|a|^3).$$

The linear term $2\hbar c|a|$ is the $k = 0$ contribution of Section 1, so the nonzero modes contribute

$$U_\perp(a) = U(a) - 2\hbar c|a| = \frac{C_M}{3\pi^2}\hbar c L|a|^2 + O(\hbar c L^2|a|^3),$$

with $C_M < 0$ for the cubic lattice (lattice-sum tables; the sign is not re-derived here and affects only the direction of a subleading correction). In the rescaled variables $a = g^{2/3}\xi/L$ this is $\frac{C_M}{3\pi^2} g^{4/3}|\xi|^2\, \hbar c/L$, of relative order $g^{2/3}$ to the zero-mode Hamiltonian $g^{2/3}h_3\, \hbar c/L$: the size that (H3) of the Feshbach reduction assumes, now with its coefficient identified.

3a. General constant backgrounds at quadratic order

The abelian computation fixes the quadratic term of $U_\perp$ for every constant background. For $|a|$ small against $2\pi/L$ each nonzero-mode quadratic form is positive definite and depends smoothly on $(a_1, a_2, a_3) \in (\mathfrak{su}(2))^3$, and the trace of the square root of a smooth positive-definite family is smooth, so with a lattice regularization $U_\perp$ is a smooth function near $a = 0$ (the non-smooth $|a|$ behaviour of U comes only from the $k = 0$ term). It is invariant under the constant gauge transformations, which rotate the three colour vectors $\vec{a}_i$ simultaneously, under the cubic group acting on the spatial index, and under $a \to -a$. The quadratic invariants of $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ under $SO(3)_{\text{colour}}$ are the bilinears $\vec{a}_i \cdot \vec{a}_j$, and the only cubic-invariant symmetric form in the spatial indices is δ_{ij}, so

$$U_\perp(a) = c\,\hbar c L \sum_{i=1}^{3} |\vec{a}_i|^2 + O(\hbar c L^2 |a|^3),$$

with a single pure number c, which the abelian valley ($\vec{a}_i = \alpha_i \hat{e}_3$, $\sum_i |\vec{a}_i|^2 = |\alpha|^2$) determines: $c = C_M/(3\pi^2)$. The Landau-level structure of non-commuting backgrounds enters only at the cubic order $O(\hbar c L^2 |a|^3)$, which is of relative order $g^{4/3}$ to the zero-mode Hamiltonian; (H3) at its leading correction therefore needs no non-abelian spectral input beyond the symmetry argument.

3b. The abelian theory in the same expansion

For compact $U(1)$ the constant modes commute, the quartic potential vanishes, and the photons are neutral, so the one-loop potential of the holonomy is identically zero: the constant-mode Hamiltonian is the free Laplacian on the torus of holonomies, $H_0 = \frac{e^2 c}{2\hbar L^3}|p|^2$ with a periodic of period $2\pi/L$, spectrum $e^2\hbar c|n|^2/(2L)$, $n \in \mathbb{Z}^3$, gap $e^2\hbar c/(2L)$. The abelian small-volume gap is linear in e^2, not $e^{2/3}$, because there is no commutator to confine the holonomy and no charged mode to give it a potential; with e fixed by the absence of running, the $1/L$ decay is uncompensated and the infinite-volume theory is gapless, in agreement with the Coulomb phase of Guth and Fröhlich–Spencer. In the non-abelian theory both the $g^{2/3}$ and the running of $g(L)$ come from the charged modes, the former through the $k = 0$ commutator term and the latter through the logarithm in the quartic coefficient of U.

4. Saturation and the centre-degenerate minima

$U(a) \le 2\hbar c \max \Phi/(\pi^2 L) = O(\hbar c/L)$ for all a, while the $k = 0$ term $2\hbar c|a|$ alone grows without bound. The linear confinement of the constant modes along the abelian valley, which in C133 holds for all $|a|$, is in the torus theory the small-La regime of a bounded periodic function. The minima of U are at $La \in 2\pi\mathbb{Z}^3$: for $SU(2)$ with the charged modes of charge ± 1, $La = 2\pi e_i$ corresponds to the fundamental holonomy $\exp(2\pi i T^3) = -1$, a centre element, trivial on all adjoint fields but a distinct state of the theory; the degenerate minima are connected by tunnelling through the valley barrier of height $O(\hbar c/L)$, which produces the splittings between the 't Hooft electric-flux sectors of the small-volume spectrum (van Baal's programme; not derived here).

Consequence for the small-volume expansion: the zero-mode ground state has width $La \sim g^{2/3}$ and energy $g^{2/3}\hbar c/L$, far below the valley saturation scale $\hbar c/L$ and the tunnelling barrier, so the C133 picture is exact at leading order and the corrections enter at relative order $g^{2/3}$, as in the Feshbach reduction. The regime where the zero-mode energy reaches $\hbar c/L$, $g(L)^{2/3} \sim 1$, is where the holonomy delocalizes over the whole compact valley, the flux sectors become degenerate and the description changes character: the crossover $z \simeq 2$ of Lüscher–Münster in this language.

5. Consequence for STATE

(H3)'s one-loop input along the abelian valley is now explicit: $U_\perp = \frac{C_M}{3\pi^2}\hbar c L|a|^2 + O(\hbar c L^2|a|^3)$, bounded by $O(\hbar c/L)$ globally, periodic with centre-degenerate minima. By the symmetry argument of Section 3a the same quadratic term holds for every constant background, so (H3) at its leading correction is a relatively bounded perturbation of h_3 by $-|C_M| g^{4/3} \sum_i |\xi_i|^2 \, \hbar c/(3\pi^2 L)$ inside the region $La \ll 1$ and by a bounded potential outside it, which the confining bound of C133 controls; the Landau-level input enters only at the next order. The abelian theory has no valley potential at all, which is the small-volume form of the abelian/non-abelian distinction.